

Md Faraz Zahir

Data Scientist | Machine Learning & MLOps

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PROFESSIONAL SUMMARY

Data Scientist with 6+ years of experience building and deploying production machine learning systems across telecom, agritech, healthcare, and retail. Delivered measurable outcomes including a 15% reduction in customer churn, 25% growth in cross-sell revenue, and a 40% reduction in excess inventory. Skilled in Python, SQL, PySpark, XGBoost, PyTorch, and MLOps tooling (MLflow, FastAPI, Docker, CI/CD), with a record of translating analytical findings into decisions for C-suite stakeholders. M.Sc. Artificial Intelligence; certified in Google Advanced Data Analytics, Stanford Machine Learning, and DeepLearning.AI Deep Learning.

EXPERIENCE

Data Scientist (Intern)

Jan 2026 – Apr 2026

Royal College of Nursing

Manchester, UK

- Build XGBoost and Random Forest classification models to forecast retention for student and early-career nurses, isolating key churn drivers across a 550,000+ member base to support the organisation's national data maturity strategy.
- Apply ARIMA forecasting and PCA to detect seasonality and regional behavioural clusters, informing targeted marketing campaigns and resource allocation.
- Own data quality assessment and cleaning of complex anonymised datasets, establishing a reliable foundation for feature importance analysis and KPI variance tracking.
- Design and deploy interactive Power BI dashboards that enable non-technical executives to track retention KPIs and evaluate strategic interventions without analyst support.
- Partner with IT and marketing teams to scope integration of ML pipelines with existing CRM systems for automated outreach triggers.

Data Scientist

Mar 2023 – Aug 2024

Corollary Kisan

India

- Developed an end-to-end customer churn prediction model in Python (scikit-learn, XGBoost, pandas) and SQL, achieving 92% ROC-AUC and cutting customer attrition by 15% within two quarters.
- Containerised and deployed a scalable ML pipeline with FastAPI and Docker, serving 1,000+ daily inferences at sub-200ms latency under CI/CD release practices.
- Implemented MLflow for experiment tracking and model versioning, reducing deployment time by 40% and making modelling workflows reproducible and auditable.
- Built predictive maintenance models on IoT time-series data using scikit-learn and PyTorch, reaching 85% accuracy and reducing critical equipment downtime by 22%.
- Engineered data and concept drift monitoring that triggered automated retraining, sustaining production model accuracy without manual intervention.
- Analysed quarterly distribution network performance in SQL to identify optimal routing points, improving SLA compliance by 20%; ran A/B tests on product placement that lifted sales by 15% at 95% confidence.

Career Break — Full-Time UPSC Civil Services Preparation

2020 – 2023

Achieved All India Rank 105, Indian Forest Service Examination

India

Analytics Consultant

Aug 2014 – Mar 2020

LinkQuest

India

- Built an ALS-based recommendation engine on 10M+ daily transactions using PySpark and Databricks, growing cross-sell revenue by 25% and improving MAP@K by 18% (0.65 hit-rate@10, validated by A/B test).
- Developed a PyTorch time-series demand forecasting model that raised forecast accuracy by 30%, reducing excess inventory by 40% and increasing stock turnover by 25%.
- Architected big data pipelines in PySpark, Databricks, and SQL processing 100GB+ daily IoT and RF sensor data, enabling real-time telecom network monitoring at 99.5% uptime; optimised SQL queries and indexing to cut extraction times by 45%.
- Analysed telecom mast data with Python (GeoPandas) and QGIS to identify coverage dead zones, driving a 10% reduction in dropped calls; mentored junior analysts and translated C-suite requirements into technical data architectures.

PROJECTS

Smart Retail Inventory & Quality Assurance System | *YOLOv9, EfficientNetB0, Transfer Learning*

- Architected a modular computer vision pipeline pairing YOLOv9 for real-time SKU localisation with EfficientNetB0 for spoilage detection, automating manual retail produce audits.
- Trained a transfer learning classifier on 29,000+ images to 98.7% accuracy on fresh vs. rotten produce; estimated 18% reduction in operational food waste.
- Tuned Non-Maximum Suppression thresholds to reach 0.52 mAP@0.5 for multi-class detection in heavily occluded shelf environments.

Real-Time Fraud Detection System | *PySpark, XGBoost, SMOTE, FastAPI, Docker*

- Engineered a streaming fraud detection pipeline scaling to 10,000+ transactions per second, achieving 93% ROC-AUC on highly imbalanced financial transaction data.
- Deployed containerised APIs with SMOTE-based class-imbalance handling, securing 89% recall on fraudulent transactions while reducing false-positive alerts for legitimate customers.

Customer Lifetime Value & Segmentation Engine | *Python, K-Means, Streamlit*

- Developed an ML engine predicting 12-month CLV by combining RFM (Recency, Frequency, Monetary) analysis with regression modelling to surface high-value segments.
- Shipped a Streamlit application enabling marketing teams to filter K-Means cohorts and export targeted outreach lists, contributing to a 12% increase in campaign conversion.

TECHNICAL SKILLS

Programming: Python, SQL (PostgreSQL, MySQL), Bash

Machine Learning & AI: scikit-learn, XGBoost, PyTorch, TensorFlow, Random Forest, Time-Series Forecasting (ARIMA), PCA, K-Means, Collaborative Filtering (ALS), Matrix Factorization, Transformers, BERT, spaCy, Hugging Face, LLMs, RAG, Computer Vision (YOLOv9, EfficientNet, Transfer Learning)

Data Engineering: PySpark, Apache Spark, Databricks, Hadoop, Apache Airflow, ETL Pipelines, Feature Engineering

MLOps & Cloud: MLflow, FastAPI, Docker, Git, CI/CD, Azure, GCP, Model Monitoring, Data Drift Detection

Analytics & BI: Power BI, Tableau, Streamlit, Matplotlib, Seaborn, QGIS, GeoPandas

Methods: A/B Testing, Hypothesis Testing, Statistical Modelling, Segmentation, Churn Modelling, Agile/Scrum, Stakeholder Engagement

EDUCATION

University of Salford

Manchester, UK

M.Sc. in Artificial Intelligence — Distinction

2026

- **Relevant Coursework:** Deep Learning, Natural Language Processing, Big Data Analytics, Computer Vision, AI Ethics.

Indian Institute of Technology, Roorkee

Roorkee, India

B.Tech in Mechanical Engineering (74%)

2014

- **Relevant Coursework:** Applied Mathematics, Operations Research, Advanced Statistics, Object-Oriented Programming.

LICENSES & CERTIFICATIONS

Google: Advanced Data Analytics Specialization

Stanford University: Machine Learning Specialization

DeepLearning.AI: Deep Learning Specialization | Unsupervised Learning, Recommenders, and Reinforcement Learning

NOTABLE ACHIEVEMENTS

UPSC Indian Forest Service Examination — All India Rank 105 (2023): placed in the top tier of one of India's most competitive national civil service examinations.

JEE Advanced — Top 1% of 1M+ candidates (2010): earned admission to IIT Roorkee.

REFERENCES

Matias Grouman – Data Strategy Lead, Royal College of Nursing

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